3-TIER LOGICAL ARCHITECTURE — DIGITAL DOCUMENT VERIFICATION SYSTEM

Software Engineering Practical Lab 09 | Architectural Decomposition, Protocols & Subsystem Contracts

TIER 1: PRESENTATION TIER (Web Client / Front-End Browser Application)

Verifier Public Portal
- Manual Verification ID Form Input (FR-22)
- Camera QR Code Decoding Interface (FR-23)
- Secure PDF Document File Upload (FR-24)

Issuing Organization Workspace
- Institutional Account Application UI (FR-07)
- Document Registration & Metadata Ingestion (FR-13)
- Issued Records Explorer & Revocation UI (FR-19)

Administrator Governance Console
- Pending Organization Review & Approval (FR-44)
- User & Document Record Governance (FR-46, FR-48)
- System-wide Verification History Explorer (FR-40)

▲ HTTPS Transport (TLS 1.3) | RESTful JSON Endpoints & Multipart Form Streams ▼

TIER 2: APPLICATION BUSINESS LOGIC TIER (Application Server)

Account Management Service
- Credential Hashing & Auth (FR-04)
- Session Lifecycle Manager (FR-11)
- Role-Based Access Enforcer (FR-09)

Document Registration Service
- File Format & Size Validator (FR-20, FR-21)
- Unique Verification ID Generator (FR-15)
- QR Code Matrix Encoder (FR-17)

Document Verification Engine
- Multi-channel Query Parser (FR-22–FR-24)
- Record Lookup Coordinator (FR-25)
- Status & Revocation Evaluator (FR-29)

Verification Audit Logger
- Guaranteed Event Recorder (FR-36, FR-37)
- Verifier Personal Ledger (FR-39)
- Multi-criteria Audit Filter (FR-42)

Cryptographic Hash Engine (IHashService)
- Standard SHA-256 Digest Computation (FR-16, Section 2.5)
- Binary Byte Stream Comparison & Tampering Detection (FR-28, FR-34)

▲ Relational Data Access Layer | SQL Queries, Transactions & Connection Pooling ▼

TIER 3: DATA PERSISTENCE TIER (Relational Database Storage)

User Store (D1)
- User & Organization Accounts (users)
- Bcrypt/Argon2 Password Hashes
- Org Status (Pending / Approved / Rejected)

Document Store (D2)
- Credential Metadata Records (documents)
- Immutable SHA-256 Registered Hashes
- Document Status (Active / Revoked) (FR-29)

Verification Log Store (D3)
- Append-Only Audit Ledger (verification_logs)
- Event Timestamp, Method & Client IP
- Outcome Verdicts & Tampering Alerts (FR-38)